Sample Midterm Exam 2

Answer Key

Econ 5030

1 Preferences

Consider an agent with preferences over $X = [0,1] \times [0,1]$ defined as follows: For all $x, x' \in X$,

$$x \succ x' \text{ if } x_1 > x_1',$$
$$x \succ x' \text{ if } x_1 = x_1' \text{ and } x_2 > x_2',$$
$$x \sim x' \text{ if } x = x'.$$

1.1 Write the definition of complete preferences and clearly demonstrate that these preferences are (or are not) complete on X.

Answer. Complete: for all $x, y \in X$, $x \succsim y$ or $y \succsim x$.

These (lexicographic) preferences **are complete**. Take any x, y. If $x_1 \neq y_1$, the bundle with the larger first coordinate is strictly preferred. If $x_1 = y_1$ and $x_2 \neq y_2$, the bundle with the larger second coordinate is strictly preferred. If $x = y$, then $x \sim y$. Every pair is ranked.

1.2 Write the definition of transitive preferences and clearly demonstrate that these preferences are (or are not) transitive on X.

Answer. Transitive: $x \succsim y$ and $y \succsim z$ imply $x \succsim z$.

Transitive. $x \succsim y$ means (x_1, x_2) is weakly above (y_1, y_2) in dictionary order: first compare x_1, and use x_2 only to break ties. Suppose $x \succsim y \succsim z$. Then $x_1 \geq y_1 \geq z_1$. If $x_1 > z_1$, then $x \succ z$. If $x_1 = z_1$, then $x_1 = y_1 = z_1$, so $x_2 \geq y_2 \geq z_2$ and $x \succsim z$.

1.3 Does a utility function exist that represents these preferences on X? Explain.

Answer. No. The preferences are complete and transitive, but they are *not continuous*. Take $x^n = (1/n, 0)$ and $y^n = (0, 1)$. Then $x^n \succ y^n$ for every n, but the limits satisfy $(0,0) \prec (0,1)$, so the preference "jumps" at the limit.

The intuition behind non-existence: each value of x_1 needs its own non-overlapping interval of utility values, $\big(u(x_1, 0), u(x_1, 1)\big)$. There are uncountably many values of $x_1 \in [0,1]$, but there are not enough rational numbers to place one inside each interval. Completeness and transitivity give a utility representation only on finite (or countable) sets. On $[0,1]^2$ we also need continuity.

1.4 Now restrict the choice set to the finite set $X' = \{0,1\} \times \{0,1\}$ with the same preferences. Does a utility function exist? If so, write one down.

Answer. **Yes.** On a finite set, complete and transitive preferences can always be represented. The ranking is $(1,1) \succ (1,0) \succ (0,1) \succ (0,0)$, so

$$u(1,1) = 4, \quad u(1,0) = 3, \quad u(0,1) = 2, \quad u(0,0) = 1$$

works (any strictly increasing relabelling also works, since utility is ordinal).

2 Game Theory

2.1 Timing and Strategies

Take a simple two player game with each player having two actions. Player 1's set of actions is $\{a_1, b_1\}$. Player 2's set of actions is $\{c_2, d_2\}$. The players' payoffs over the four possible action combinations are as follows:

$$\begin{aligned} u_1(a_1, c_2) &= 20 & \qquad u_2(a_1, c_2) &= 10 \\ u_1(a_1, d_2) &= 5 & \qquad u_2(a_1, d_2) &= 0 \\ u_1(b_1, c_2) &= 15 & \qquad u_2(b_1, c_2) &= 5 \\ u_1(b_1, d_2) &= 30 & \qquad u_2(b_1, d_2) &= 15 \end{aligned}$$

Suppose this is a sequential choice game in which player 1 goes first, then player 2 observes player 1's choice and makes her choice.

2.1 Write the Normal Form of the game.

Answer. $N = \{1,2\}$, $S_1 = \{a_1, b_1\}$, $S_2 = \{c_2c_2, c_2d_2, d_2c_2, d_2d_2\}$, where xy means "x after a_1, y after b_1." $u_i(s_1, s_2)$ is the payoff at the terminal node reached by (s_1, s_2).

2.2 Construct the Game matrix.

Answer.

	c_2c_2	c_2d_2	d_2c_2	d_2d_2
a_1	20, 10	20, 10	5, 0	5, 0
b_1	15, 5	30, 15	15, 5	30, 15

2.3 Find the set of strategies that remain after iterated deletion of strictly dominated strategies.

Answer. P1: $a_1 = (20, 20, 5, 5)$ and $b_1 = (15, 30, 15, 30)$; neither dominates. P2 (payoffs vs a_1, b_1): $c_2c_2 = (10, 5)$, $c_2d_2 = (10, 15)$, $d_2c_2 = (0, 5)$, $d_2d_2 = (0, 15)$. c_2d_2 strictly dominates d_2c_2 ($10 > 0$, $15 > 5$), so delete d_2c_2. The other comparisons have a tie, so there is only weak dominance. Next round, P1 vs $\{c_2c_2, c_2d_2, d_2d_2\}$: $a_1 = (20, 20, 5)$ and $b_1 = (15, 30, 30)$, so there is still no dominance.

Surviving set: $\{a_1, b_1\} \times \{c_2c_2, c_2d_2, d_2d_2\}$.

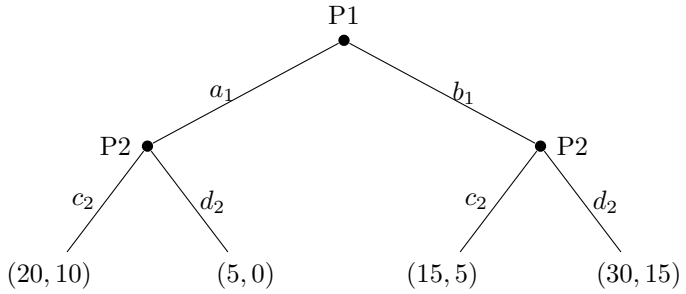
2.4 Find the set of Nash Equilibrium.

Answer. P1 best responses: a_1, b_1, b_1, b_1 against the four columns. P2 best responses: vs a_1, $\{c_2c_2, c_2d_2\}$; vs b_1, $\{c_2d_2, d_2d_2\}$.

$$NE = \{(a_1, c_2c_2),\ (b_1, c_2d_2),\ (b_1, d_2d_2)\}.$$

2.5 Write the Game Tree and find the set of Subgame perfect equilibrium. Which Nash equilibria are based on incredible threats?

Answer.



After a_1, P2 plays c_2 ($10 > 0$); after b_1, P2 plays d_2 ($15 > 5$). P1 then compares 20 with 30 and chooses b_1. $SPE = \{(b_1, c_2d_2)\}$.

(a_1, c_2c_2) relies on the incredible threat to play c_2 after b_1 (P2 would actually get $15 > 5$ from d_2). This threat changes the outcome. (b_1, d_2d_2) has the SPE outcome, but its off-path plan (d_2 after a_1) is not credible, so it is not subgame perfect.

2.2 Games with Different Preferences

There are two players and each player i has two actions $\{R_i, Q_i\}$. The money each player i receives for the different action pairs (in a game with player j) is described in the matrix below (*this is not the Game Matrix*):

		$P2$	
		R_2	Q_2
$P1$	R_1	\$8, \$8	\$2, \$12
	Q_1	\$12, \$2	\$0, \$0

The game is sequential. Player 1 picks her action first and then player 2 observes the action of player 1 and player 2 picks his action. We will consider 3 cases of this game.

Case 1: Both players only care about their own money with more money being strictly better than less.

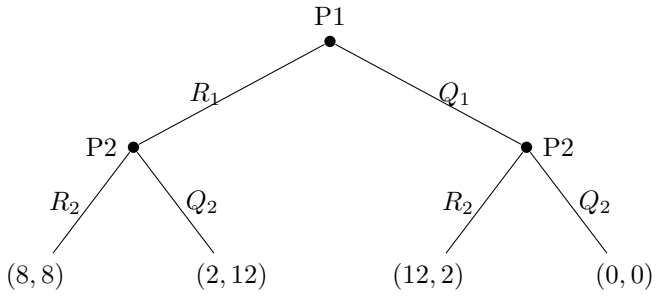
2.6 Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

Answer. $S_1 = \{R_1, Q_1\}$; $S_2 = \{R_2R_2, R_2Q_2, Q_2R_2, Q_2Q_2\}$ (action after R_1, action after Q_1).

	R_2R_2	R_2Q_2	Q_2R_2	Q_2Q_2
R_1	8, 8	8, 8	2, 12	2, 12
Q_1	12, 2	0, 0	12, 2	0, 0

P1 best responses: Q_1, R_1, Q_1, R_1. P2 best responses: vs R_1, $\{Q_2R_2, Q_2Q_2\}$; vs Q_1, $\{R_2R_2, Q_2R_2\}$.

$$NE = \{(Q_1, R_2R_2),\ (Q_1, Q_2R_2),\ (R_1, Q_2Q_2)\}.$$



Backward induction: after R_1, P2 plays Q_2; after Q_1, P2 plays R_2. P1 compares 2 with 12 and plays Q_1. $SPE = \{(Q_1, Q_2R_2)\}$. (This is a sequential game of Chicken: moving first is an advantage.) (R_1, Q_2Q_2) is P2's incredible threat to "crash" after Q_1.

Case 2: Each player only cares about the sum of the two players' money and strictly prefers more total money to less.

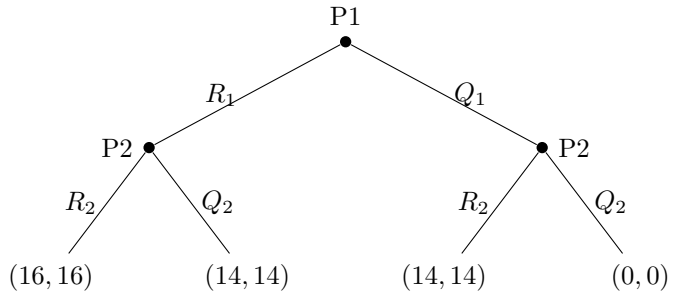
2.7 Create a normal form for this game. Write the game matrix and find the set of Nash Equilibrium. Draw the game tree and find the set of Subgame perfect equilibrium.

Answer. Common payoff = total money: $16, 14, 14, 0$.

	R_2R_2	R_2Q_2	Q_2R_2	Q_2Q_2
R_1	16, 16	16, 16	14, 14	14, 14
Q_1	14, 14	0, 0	14, 14	0, 0

P1 best responses: R_1, R_1, $\{R_1, Q_1\}$, R_1. P2 best responses: vs R_1, $\{R_2R_2, R_2Q_2\}$; vs Q_1, $\{R_2R_2, Q_2R_2\}$.

$$NE = \{(R_1, R_2R_2),\ (R_1, R_2Q_2),\ (Q_1, Q_2R_2)\}.$$



P2 plays R_2 after either action, and P1 plays R_1 ($16 > 14$). $SPE = \{(R_1, R_2R_2)\}$.

Case 3: Player 1 only cares about her own money. Player 1 does not know Player 2's type. With probability $\mu \in (0,1)$, Player 2 is *spiteful*: his payoff is his own money minus Player 1's money. With probability $1-\mu$, he is selfish (cares only about his own money). Player 2 knows his own type.

2.8 Create a game tree for this game and use a form of backward induction as your solution concept. For which values of μ does Player 1 choose Q_1? Explain clearly the difficulties you faced in finding a way to analyze the game.

Answer. *Tree:* Nature picks spiteful (μ) or selfish ($1-\mu$). P1 moves without seeing this, so her two nodes form one information set. Each type of P2 observes P1's action and then moves.

Spiteful payoffs (own − other's): $(R_1, R_2) \to 0$, $(R_1, Q_2) \to 10$, $(Q_1, R_2) \to -10$, $(Q_1, Q_2) \to 0$.

	after R_1	after Q_1
Spiteful P2	Q_2 ($10 > 0$)	Q_2 ($0 > -10$)
Selfish P2	Q_2 ($12 > 8$)	R_2 ($2 > 0$)

$E[R_1] = 2$ (both types play Q_2). $E[Q_1] = \mu \cdot 0 + (1-\mu) \cdot 12 = 12(1-\mu)$.

So P1 plays Q_1 iff $12(1-\mu) > 2 \iff \mu < 5/6$, plays R_1 if $\mu > 5/6$, and is indifferent at $\mu = 5/6$.

Difficulties: P1's decision cannot be solved by plain backward induction, because she does not know which subgame she is in. We need to add a Nature move, an information set, and beliefs, and then maximize expected payoff. We also have to turn money into payoffs for each type. The spiteful type is willing to destroy money (choose $(\$0, \$0)$) just to hurt P1. The prediction hinges on μ, which is not observable.

3 Conceptual Questions

3.1 What is expected utility? What extra assumption (beyond complete, transitive and continuous preferences over lotteries) is required for an expected utility representation? Give an example of behavior that violates it.

Answer. Preferences over lotteries $L = (p_1, \ldots, p_K)$ on outcomes $z_1, \ldots, z_K$ have an *expected utility* representation if there is a utility index u on outcomes such that $L \succsim L' \iff \sum_k p_k u(z_k) \geq \sum_k p'_k u(z_k)$. The value of a lottery is the probability-weighted average of the utilities of its outcomes, which is linear in the probabilities.

The extra assumption is the **independence axiom**: $L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''$ for all L'' and $\alpha \in (0,1)$. Mixing in a common lottery must not change the ranking. This is exactly the linearity in probabilities that EU needs.

Violation (Allais). Many people choose \$1M for sure over the gamble {10%: \$5M, 89%: \$1M, 1%: \$0}. The same people choose {10%: \$5M, 90%: \$0} over {11%: \$1M, 89%:

\$0}. Both pairs differ only by a common 89% component (\$1M in the first pair, \$0 in the second). Independence says the ranking cannot flip, but it does.

3.2 Define a best response and a (pure strategy) Nash Equilibrium. What does Nash Equilibrium assume that rationalizability does not?

Answer. Best response: $BR_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i})$. A profile s^* is a *Nash equilibrium* if $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$ for every $s_i \in S_i$ and every player i. In other words, $s_i^* \in BR_i(s_{-i}^*)$ for all i, so no player has a profitable *unilateral* deviation.

Rationalizability assumes only rationality and common knowledge of rationality: each strategy must be a best response to *some* belief that is itself consistent with rationality. NE adds *correct beliefs*: each player best responds to what the others *actually* play. That consistency is why NE is a subset of the rationalizable strategies and usually predicts much more sharply.

3.3 In a sequential game, what is a strategy? Why does the second mover have more strategies in the sequential version of a game than in the simultaneous version, even though she has the same actions?

Answer. A strategy is a *complete contingent plan*: it specifies an action at every decision node (information set) where the player might move, including nodes that are never reached. In the simultaneous game, P2 has one information set, so $S_2 =$ her action set (2 strategies). In the sequential game, P2 observes P1's action and has one node per P1 action. She must plan for each, so $|S_2| = 2 \times 2 = 4$. Actions describe what you physically do; strategies describe what you would do in every contingency. Only the strategy sets differ between the two normal forms.

3.4 Define consumer surplus, producer surplus and total surplus in a market game. Why is total surplus a natural starting point for evaluating welfare, and why is it not a complete answer?

Answer. $CS = \sum(WTP - p)$ over units bought: the area under demand and above price. $PS = \sum(p - WTS)$ (or $p - MC$) over units sold: the area above supply/MC and below price. $TS = CS + PS$, the total gains from trade.

TS is natural because it measures how much value the market creates. Maximizing it is equivalent to Pareto efficiency when transfers are possible, and losses show up as deadweight loss. It is not complete because it is about *efficiency*, not *equity*. A dollar of surplus counts the same whether it goes to a poor consumer or a rich firm owner, so TS is silent on how the gains are distributed. Any notion of equity requires an extra normative criterion.

4 Cournot Oligopoly

Consider a Cournot oligopoly with 3 firms. Firms simultaneously choose quantities $q_i \geq 0$ of a homogeneous good. Each firm has cost $c(q_i) = 10q_i$. Market inverse demand is $P(Q) = 100 - Q$, where $Q = q_1 + q_2 + q_3$ and $Q_{-i} = Q - q_i$.

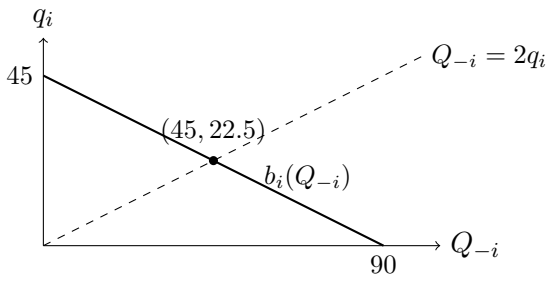
4.1 Write the Normal Form of this game.

Answer. $N = \{1, 2, 3\}$; $S_i = [0, 100]$; $\pi_i(q_i, Q_{-i}) = (100 - q_i - Q_{-i})q_i - 10q_i$.

4.2 Find the best response function of firm i, $b_i(Q_{-i})$, and graph it on the Q_{-i}–q_i plane.

Answer. π_i is quadratic and concave in q_i ($\partial^2\pi_i/\partial q_i^2 = -2 < 0$). The FOC is

$$100 - 2q_i - Q_{-i} - 10 = 0 \implies b_i(Q_{-i}) = \frac{90 - Q_{-i}}{2} \quad \text{(and 0 if } Q_{-i} > 90\text{)}.$$



The symmetric equilibrium is where the best response crosses the symmetry line $Q_{-i} = 2q_i$.

4.3 Find the Cournot Nash equilibrium quantity of each firm q^c, total output Q^c and price p^c.

Answer. Impose symmetry *after* the FOC: $q^c = (90 - 2q^c)/2 \Rightarrow 4q^c = 90 \Rightarrow q^c = 22.5$. Then $Q^c = 67.5$ and $p^c = 100 - 67.5 = 32.5$.

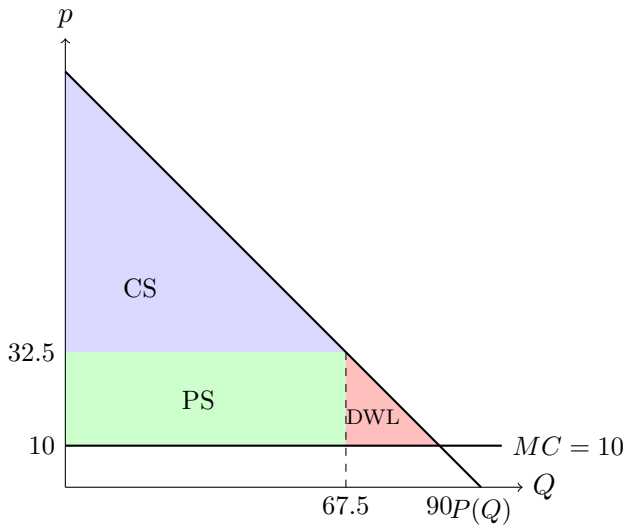
4.4 Find each firm's profit, CS, PS, and TS.

Answer. $\pi_i = (32.5 - 10)(22.5) = 506.25$. $PS = 3\pi_i = 1518.75$. $CS = \frac{1}{2}(67.5)(100 - 32.5) = 2278.125$. $TS = 3796.875$.

4.5 Find the efficient outcome. Define and compute the Dead Weight Loss. Draw the market on the p–Q plane showing CS, PS and DWL.

Answer. Efficient: $p = MC = 10 \Rightarrow Q^* = 90$, $TS^* = \frac{1}{2}(90)(90) = 4050$.

Deadweight loss is the total surplus lost relative to the efficient outcome, from trades worth more to buyers than they cost to produce that do not happen: $DWL = TS^* - TS^c = \frac{1}{2}(90 - 67.5)(32.5 - 10) = 253.125$.



4.6 Suppose the three firms form a cartel that maximizes joint profit and splits output equally. Find the cartel output and each firm's profit. Is this a Nash equilibrium? Explain.

Answer. Joint profit $(100 - Q)Q - 10Q$ is maximized at $90 - 2Q = 0$, so $Q^m = 45$, $p^m = 55$, and total profit is $45^2 = 2025$. Each firm produces 15 and earns $675 > 506.25$.

This is **not** a NE. Holding the other two at 15 each ($Q_{-i} = 30$), firm i's best response is $b_i(30) = (90 - 30)/2 = 30 \neq 15$. Deviating gives $Q = 60$, $p = 40$, and $\pi_i = 30 \cdot 30 = 900 > 675$. Each firm ignores that its extra output lowers its rivals' profit, exactly as in the Prisoner's Dilemma. The cartel outcome is better for the firms but unstable. (It is also worse for society: its DWL is bigger.)

4.7 Generalize to n firms with the same demand and costs. Find $q^c(n)$, $p^c(n)$ and $DWL(n)$. What happens as n grows? What does this say about the effect of competition on efficiency?

Answer. With $Q_{-i} = (n-1)q$: $q = (90-(n-1)q)/2 \Rightarrow q^c(n) = \dfrac{90}{n+1}$, $Q^c(n) = \dfrac{90n}{n+1}$, $p^c(n) = \dfrac{100+10n}{n+1}$.

$$DWL(n) = \tfrac{1}{2}\Big(90 - \tfrac{90n}{n+1}\Big)\Big(p^c(n) - 10\Big) = \frac{90^2}{2(n+1)^2} = \frac{4050}{(n+1)^2}.$$

Check: $n = 1$ (monopoly) gives $Q = 45$, $p = 55$; $n = 3$ gives $DWL = 253.125$. As $n \to \infty$, $p^c \to 10 = MC$, $Q^c \to 90$ and $DWL \to 0$. More firms means each firm's output decision has a smaller effect on price, so the underproduction distortion shrinks toward the competitive, efficient outcome.